# MUHAMMAD ABUBAKAR

**AI / Machine Learning Engineer**

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## PROFESSIONAL SUMMARY

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AI/ML Engineer with hands-on experience in Python, deep learning, and computer vision, including transfer learning with ResNet34 and EfficientNetB0, attention mechanisms (CBAM), and classical machine learning (SVM, Random Forest, Logistic Regression). Experienced across the full ML workflow: data preprocessing, feature engineering, model training, and evaluation (precision, recall, F1-score, ROC-AUC). Built a client-facing computer vision system achieving 96.46% test accuracy, and combines this with REST API and full-stack development experience to build and ship practical ML-powered applications. Experienced in applying machine learning and computer vision techniques to classification and prediction problems, with practical exposure to model comparison, feature extraction, data augmentation, and performance evaluation.

## TECHNICAL SKILLS

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**Programming:** Python, JavaScript, C++, C, Assembly (EMU 8086)

**Machine Learning:** Support Vector Machines (SVM), Random Forest, Logistic Regression, XGBoost, Transfer Learning

**Deep Learning:** PyTorch/TensorFlow workflows, ResNet34, EfficientNetB0, CBAM Attention Module

**Computer Vision:** Image Classification, Image Processing, Data Augmentation

**Data & Evaluation:** Data Preprocessing, Feature Engineering, Exploratory Data Analysis, Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix

**Frameworks & Libraries:** Scikit-learn, Pandas, NumPy, Matplotlib

**Software Engineering:** REST API Design, MERN Stack (MongoDB, Express.js, React.js, Node.js), Tailwind CSS, Authentication & Session Management

**Tools & Platforms:** Google Colab, Jupyter Notebook, VS Code, Postman, Git/GitHub

## MACHINE LEARNING / AI EXPERIENCE

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**Machine Learning Intern — CodeAlpha** — Remote | Nov 2025 – Jan 2026

**Credit Scoring & Loan Default Prediction** | Python, Pandas, Scikit-learn, Random Forest

- Developed and evaluated a credit risk classification model using Random Forest on historical financial data, applying data cleaning, missing-value treatment, and feature engineering.
- Evaluated model performance using precision, recall, F1-score, confusion matrix, and ROC-AUC to assess loan default prediction reliability.

**Disease Prediction System** | Python, Scikit-learn, Logistic Regression, SVM, Random Forest

- Built and compared Logistic Regression, SVM, and Random Forest classifiers to predict disease risk from patient diagnostic attributes, applying feature scaling and preprocessing.
- Conducted comparative model evaluation across three algorithms to identify the most effective model for the dataset.

**Sales Prediction Using Machine Learning** | Python, Pandas, NumPy, Matplotlib, Scikit-learn

- Developed a regression model using Pandas, NumPy, and Scikit-learn to predict product sales from advertising and historical sales data.
- Visualized trends with Matplotlib to communicate sales drivers and model insights.

## SELECTED AI/ML PROJECTS

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**AI-Based Student Engagement Detection System — Client Project** | EfficientNetB0, XGBoost, TensorFlow | May 2026 – Jun 2026

- Developed an AI-powered system classifying six classroom engagement states (Focused, Confused, Frustrated, Bored, Looking Away, Drowsy) using deep learning and gradient boosting.
- Implemented transfer learning and deep feature extraction with EfficientNetB0, followed by XGBoost-based classification, achieving 98.76% training accuracy and 96.46% testing accuracy.
- Evaluated performance using confusion matrix, precision, recall, and F1-score across all six engagement classes.

• Designed the solution for potential webcam-based real-time classroom monitoring, providing a foundation for automated engagement analysis.

**Plant Leaf Disease Classification Using Deep Learning** | ResNet34, EfficientNetB0, CBAM

• Built a multi-class plant disease classifier on the Plant Village dataset, implementing and comparing ResNet34, EfficientNetB0, and EfficientNetB0 with CBAM attention.

• Applied image preprocessing, data augmentation, and model optimization; the attention-based EfficientNetB0 model achieved the highest classification accuracy among the three architectures tested.

**Brain Tumor Detection Using Machine Learning** | SVM Classifier

• Built an MRI-based brain tumor detection model using an SVM classifier, preprocessing images and extracting key features to distinguish tumor from non-tumor cases.

• Strengthened experience in medical image processing, feature extraction, and classification evaluation.

## SOFTWARE ENGINEERING EXPERIENCE

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**Management Trainee Officer (MTO) – Information Technology** | Transsion Holdings (Infinix, TECNO & itel) | Jul 2026 – Present

• Support IT operations, troubleshooting hardware, software, and network issues, and assisting with system administration and IT asset management across departments.

• Collaborate with senior IT professionals to implement and maintain enterprise IT infrastructure.

**Full-Stack Developer – Help Desk Ticketing Portal** | Independent Project | React, Vite, Tailwind CSS, Node.js, REST API

• Designed and built a full-stack IT help desk/ticketing application using React, Vite, and Tailwind CSS with a custom Node.js REST API backend.

• Implemented complete ticket lifecycle management with role-based access for employees, engineers, and admins.

• Implemented token-based session authentication and salted-hash password storage for secure access control.

**Full-Stack Developer – Gift Inventory Management System** | MERN Stack | Independent Project |

- Developed and deployed a full-stack gift inventory management web application to manage promoter records, gift inventory, and associated data through a centralized platform.
- Implemented promoter login functionality and image-upload capabilities to support inventory tracking and maintain organized records.
- Integrated MongoDB Atlas for cloud database management and deployed the application on Vercel for online accessibility.

**Web Development Intern — Hex Softwares** | Remote | Feb 2025 – Apr 2025

• Built responsive e-commerce, real estate listing, and task management web applications using HTML, CSS, JavaScript, and Bootstrap.

**Additional Software Projects**

• Intelligent Voting Platform with AI Integration (MERN Stack): built a secure full-stack digital voting platform integrating multilayer biometric security (CNIC validation, facial recognition, fingerprint scanning) and OTP-based multi-factor authentication.

• Hospital Management System (React, MongoDB): built a full-stack application with patient records, doctor profiles, appointment scheduling, and real-time updates.

## EDUCATION

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**BS in Computer Science — University of Management and Technology (UMT), Lahore** | 2022 – 2026

• Relevant coursework: Artificial Intelligence, Data Science, Data Structures & Algorithms, Databases, Operating Systems, Object-Oriented Programming.

## CERTIFICATIONS

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• Machine Learning Internship – CodeAlpha

• Python

• Web Development Internship – Hex Softwares

• React Development

## LEADERSHIP / ACTIVITIES

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**Orientation Leader — UMT Fall Batch 2024**

• Led campus orientation programs to help new BS students settle into university life; organized community activities.

## LANGUAGES

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Fluent in Urdu, Punjabi, and English.